Name: ______________________ Andrew Id: ______________________

15-112 Spring 2026 Exam 1 Review

Up to 40 minutes. No calculators, no notes, no books, no computers. Show your work!
Do not use lists, tuples, dictionaries, sets, try/except, or recursion on this review.

1. (points) **Code Tracing**:

```
def ct1(s):
    t = ''
    for i in range(len(s)):
        t += s[i:] + s[i:][::-1]
    for c in s[::-1]:
        t = (t.replace(c+c,'') + str(t.count(c) + 1))
    print(t)


print(ct1('abcd'))
```

2. (points) **Code Tracing**:

```
def ct2(n):
    def f(n):
        r,m = 0,n
        while m>0:
            r,m = r+m,m-1
        return 2*r-n
    def g(n):
        return (f(n+1) - f(n-1))
    return f(g(n))

print(ct2(3))
```

3. (points) **Free Response**: largest(s)

   Write a function `largest(s)` which takes a text and returns the largest number in the string. For example largest("I have 3 courses, but one of the courses have 20 credits. I don't know how to score 100 in that course") should return 100.

4. (points) **Free Response**:

A number is termed Witty if the number begins or ends with an even number, but not both. 2 is a not a Witty number, but 21 is. Create a function `nthWittyCompositeNumber(n)` to find the nth witty number which is also a composite. *Remember: A composite number is: a positive integer greater than 1 that has divisors other than 1 and itself.*
Use helper functions wherever necessary.

5. (points) **Free Response**: wordsEndingInNumber(s)

Write the function `wordsEndingInNumber(s)` which, given a string s, returns a string with a comma-separated list of words from the string that end with a number. You may assume that words in the string are separated by spaces.

For example: `wordsEndingInNumber("purple monkey5 dish washer12 74 2evergreen")` returns `"monkey5,washer12,74"`.